

BHABHA UNIVERSITY, BHOPAL

NH-12, Hoshangabad Road, Jatkhedhi, Bhopal-462026

BACHELOR OF COMPUTER APPLICATIONS (BCA)



**DETAILED SYLLABUS
(EFFECTIVE FROM JULY 2018)**

**Department of Computer Science & Application
BHABHA UNIVERSITY, BHOPAL**

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DETAILED SYLLABUS
BCA THIRD YEAR

S.No.	Paper Code	Paper Name	Internal Evaluation	Theory	Practical	Total
1	BCA31	Data Communication and Networking	25	50		75
2	BCA32	Web Design	25	50		75
3	BCA33	Organization Behavior	25	50		75
4	BCA34	Managerial Economics and MIS	25	50		75
5	BCA35	System analysis and Design	25	50		75
6	BCA36	DESK TOP PUBLISHING	25	50		75
Practical						
7	BCA37	DTP with HTML	20		30	50
8	BCA38	Project	50		50	100
Total			220	300	80	600

BCA31 DATA COMMUNICATION AND NETWORKING

UNIT-I

Data communication and IT, Communication Mode- Simplex, Half Duplex, Full Duplex, Communication Channels - Twisted, Coaxial, Fiber Optic, Serial and Parallel Communication, Types of Network - LAN, WAN, MAN ,Internet etc.

UNIT-II

Topologies of LAN - Ring, Bus, Star, Mesh and Tree topologies, World Wide Web Internet Services, Analog & Digital Signal. Base Band , Broad Band, Multiplexer FDM, TDM, Modulation AM, FM, PM, Transmission Media ,Modem. OSI Reference Model , Switching Technique, Message Switching, Circuit Switching, Packet Switching, Virtual Circuit

UNIT-III

Fast Ethernet, FDDI Token Ring, Wireless LAN, Inter-Networking Devices, Bridge, Routers Gateways, Repeater, Distance Vector Routing, Shortest Path Routing, Broadcast Routing, Multicast Routing, TCP/IP Protocol.

UNIT-IV

Comparison between OSI and TCP/IP Models, TELNET, FTP, SMTP, MINE, UDP, URL (Uniform Resource Locator) HTTP differentiated services, network interconnection model, MPLS, ATM applications.

UNIT-V

Network Security : Network Security Issues, Firewalls – Need and Features of Firewalls, Types of Firewall Technology- Network Level and Application Level, Cryptography, Cryptography algorithms, security protocol

REFERENCE BOOKS :

- 1•DATA COMMUNICATION BY PRAKASH C GPTA
- 2•COMPUTER NETWORKING BY ANDREWS TANANBAUM
- 3• UNDERSTANDING DATA COMMUNICATION OF NETWORKING BY WILLIAM A SHAY
- 4• COMMUNICATION AND NETWORK BY LEWIS MACHENZIE

BCA32 Web Design

UNIT - I

Internet & World Wide Web: Introduction, Internet Addressing, Browsers, URL, Web Server, Web Site, Homepage, Domain Names Basic Concepts. Facilities available over the Internet – email, www, ftp, telnet, usenet, blog, chat. Components of Internet – client/server modem, cable, modem, backbone, router, IP addresses, address classes, DNS names, Introduction to HTML overview, Structure of HTML documents, Types of Documents, HTML Elements and Attributes.

UNIT - II

Basic Formatting of HTML documents layout – Font, Lists, Paragraph, Break Rule, Horizontal Rule, Colors, Backgrounds, text elements, etc. Links in html anchor element and its attributes, images and anchors, using META information. HTML Media Types, Audio & Video support in web browsers.

UNIT - III

Advanced Layout: Tables, Frames, Layers, Tables layout, elements and attributes, ROWSPAN, COLSPAN. Frames using frames, layout of frames, problems using frames. Layers-concept of layers, positioned and inflow layers.

UNIT - IV

Style sheets – Basic concept of style sheet, using style sheet, cascading style sheets (css), using style sheets – basics and properties, positioning with style sheets, Basic Interactivity in html: Forms- Concept of Forms, <FORM> element, attributes, Controls used for forms, examples of form design

UNIT - V

Advanced Features in HTML: Scripting, DHTML & XML, Web publishing – Concepts. Scripting – purpose of scripting, specifying scripts & designing events. DHTML: Overviews of DHTML and Document object Model, html & scripting access, rollover buttons XML: Basic XML structure, ways to use XML, rewriting html as xml.

REFERENCE BOOKS:

1. The Complete Reference to HTML – by Thomas A Powell, TMH, II Edition.
2. Using HTML – by Lee Anne Phillips, PHI
3. Learning computer Fundamentals, MS Office & Internet & web technology by Dinesh Maidasani, Firewall Media.
4. Introductions to Internet & Java – by Kiran Nayar, Sukhjeet Kaur, Kalyani Publishers.

BCA33 - ORGANIZATION BEHAVIOR

UNIT I

ORGANISATIONAL What is Psychology, Whom does the Psychologist observe, What does Psychologist observe, Where does the Psychologist observe,

UNIT II

Sensation & Perception, Feelings and Emotions, Learning and Thinking.(in so far as they are applicable to Industry), Personality, Definition & Theories, Assessing Personality.

UNIT III

An Overview of Transactional Analysis as a Tool for Measuring One's Own Awareness and of helping Interpersonal Relationship in an Organization. Theories of Organization, Contingency Model and other Models of Organization, Theories of Motivation. – Abraham Maslow, Herzberg Mclelland.

UNIT IV

Group Dynamics, Structure, Process, Values of Groups, Role and Stature. Theories of Leadership, Identifying Leadership Potential.

UNIT V

Communication: Process, Methods, Barriers and Usefulness.

TEXT BOOKS:

1. Organizational Behavior – by Robbins.
2. Organizational Behavior – by Luthans.
3. Organizational Behavior – by Sashi Gupta & Rozy Joshi.

BCA34 - MANAGERIAL ECONOMICS AND MANAGEMENT INFORMATION SYSTEM

Unit I

Managerial Economics: Introduction of Managerial Economics- Factors Influencing Manager, Micro and Macroeconomics, Theory of the Cost Theory of the Firm Theory of Production Function, Production System.

Unit II

Input output Analysis, Micro Economics Applied to plants and industrial undertakings, Productivity, Factors affecting Productivity, Increasing Productivity of Resources

Unit -III

Meaning, Nature, Need, Role, Importance, Evolution of Management Through Information system. Structure of Management Information System. Relatedness of MIS with management activities, Management functions and decision making, Information System in Business and Management.

Unit-IV

Development of MIS- Methodology and Tools/ Techniques for systematic designing, implementation, evaluation. modification of MIS, A study of major financial, production, Man power and marketing MIS and case Studies.

Unit -V

Advanced MIS-concept need and problems in achieving advanced MIS, Decision support System. Rationale of computer application, Decision support system (DSS).

Reference Books:

1. Managerial Economics: Joel Dean: Prentice Hall of India
2. Murdick, RG, Ross, J.E. & Claggett, J.R.: Information systems for Modern Management, PHI
3. Thomas, R. & Prince: Information systems for planning and control.
4. Aktas ; structured analysis and design of information system, PHI
5. Spargue and Watson : Decision Support System ,2nd Edn. prentice Hall international ggg

Unit-I

System Concept: Definition, Characteristics, Elements of system, Physical and abstract System open and closed system, man-made information systems, System Development Life Cycle: Various phases of system development, Considerations for system planning and control for system success. System Planning

Unit-II

Tools of structured Analysis: Logical and Physical models context, diagram, data dictionary ,data diagram, form driven methodology, IPO and HIPO charts, Gantt charts, system model, pseudo codes, Flow charts, system flow chart, run flow charts etc., decision tree, decision tables, data validation. Management

Unit-III

System testing & quality: System testing and quality assurance, steps in system implementation and software maintenance. System security: Data Security, Disaster/ recovery and ethics in system development, threat and risk analysis, System audit. System Analyst

Unit-IV

The Software Product and Software Process: Software Engineering - A Layered Technology, Software Process Models: Linear Sequential Model. Prototyping Model, RAD Model Evolutionary Software Procsss Models: Incremental Model, Spiral Model Component Assembly Model, Systems Engineering: The Systems Engineering Hierarchy, Information Engineering, Information Strategy Planning, Business Area Analysis, Product Engineering Requirement Analysis Modeling: Analysis Concepts and Principles, The Elements of the Analysis Model Data Modirying, Functional Modeling and Information Flow and Behavior Modeling, Mechanics of Structured Analysis, Data Dictionary.

Unit-V

Principles and Methods: The Software Design Process: Design Principles, Design Concepts, Effective Modular Design, Design Heuristics, Design Documentation, Design Methods: Data Design, Architectural Design, Interface Design, Human Computer Interface Design, Procedural Design.

REFERENCE BOOKS:

1. Computers Today – by Suresh K Basandra.
2. Modern System Analysis & Design by A Hoffer, F George, S Valaciah, Low price Edn. Pearson Education.
3. Introduction to Systems Analysis and Design – by Lee.

DESK TOP PUBLISHING: Why Graphics, Various Types of Graphics Programs. Drafting Packages, Introductions to Printing, DTP Package, Advantage of D.T.P in Publication, Microsoft Windows. Various Documentation cum DTP Package e.g. Word Perfect, Microsoft Word Etc.

Unit II

Introduction To PageMaker / Ventura or a similar Package. Page Maker Icon and help, Tool Box, Styles, Menus etc., Different screen Views, Importing text/Pictures Preparation of Document using DTP Package.

Unit III

Text Formatting, Different Page Layouts, Printing Various Fonts And Characters Sets. Various Type of Printers used in DTP.

Unit IV

Introduction to Commercial DTP System available in Market, Indian Language Fonts, Creation of Indian Language Fonts.

Unit V

Import & Export of Documents created by other Word Processors, Spelling Check, Designing exercise like Visiting Card, Greeting Cards etc.

Reference Books

Desktop Publishing on PC by MC Shrama, BPB Publication.

PAGE MAKER 4.0 & 5.0 BY B.P.O. PUBLICATIONS1. PAGEMAKER – MANUAL 2. „O“ LEVEL MODULE M3.2 DESKTOP PUBLISHING & PRESENTATION GRAPHICS BY V K JAIN, BPB PUBLICATIONS.